
ROBOT RIOT

2D Top-down Roguelike Action

Final Presentation

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1. How to Play

Controls & Game Loop

Controls

WASD	Move
Left Click	Basic Attack (Melee / Laser)
Right Click	Special Attack
Shift	Dash (i-frames)
Q	Character Skill
E	Pick Up Weapon
Tab	Swap Weapon Slot
ESC	Pause / Return to Lobby

Game Loop

- Pick a character in the Lobby (GPT / Claude / Gemini)
- Progress through Floor 1 → 2 → 3
- Clear all 5 rooms on a floor to summon the boss
- The last room you enter becomes the boss room
- Defeat the boss → portal appears → next floor
- On death or clear, return to lobby — coins fund permanent upgrades

2. Implementation

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Code Structure

Project layout (Assets/Scripts)

- Combat/ Weapons, items, coin, skill
- WeaponController, LaserWeapon, SkillController, Coin
- Entities/ Player, enemies, bosses, projectiles
- PlayerController, BaseEntity, HumanAI, MadRushBoss
- Managers/ UI, game state, lobby, input, persistence
- GameManager, UIManager, PauseMenu, KeyBindings, PermanentData
- Map/ Procedural rooms, floors, portal
- RoomController, RoomManager, FloorInitializer(1/2/3), Portal

2. Implementation — Core Logic 1

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Weapon System: WeaponController + LaserWeapon

A single controller branches between melee and laser based on `WeaponData.weaponType`.

```
// WeaponController.Update() – branch by weapon type
if (currentWeaponData.weaponType == WeaponType.Laser)
{
    bool firing = Mouse.current.leftButton.isPressed;
    if (laserWeapon != null) laserWeapon.SetActive(firing);
    return;
}

// Melee: left-click swings, right-click special
if (Mouse.current.leftButton.wasPressedThisFrame && Time.time >= nextAttackTime)
    StartCoroutine(PerformAttack());
if (Mouse.current.rightButton.wasPressedThisFrame && Time.time >= nextSpecialAttackTime)
    StartCoroutine(PerformSpecialAttack());
```

Highlights

- Swap one `WeaponData` (`ScriptableObject`) → stats, trail, special, cooldown all update
- Laser uses `LineRenderer` auto-aim beam; melee uses fan-shaped `OverlapCircleAll` hit detection
- Combo counter + hit-stop + camera shake for satisfying feedback

2. Implementation — Core Logic 2

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Dynamic Boss Placement: RoomManager.TryClaimBossFor

Instead of fixing a boss room, the last room the player enters becomes the boss room.

```
// RoomManager.TryClaimBossFor() – place boss in the last entered room
public GameObject TryClaimBossFor(RoomController room)
{
    if (pendingBossPrefab == null || claimedBossRoom != null) return null;

    int remaining = 0;
    foreach (var r in rooms)
        if (r != room && !r.IsCleared) remaining++;

    if (remaining > 0) return null;    // other rooms still uncleared → no boss yet
    claimedBossRoom = room;
    return pendingBossPrefab;        // spawn the boss here
}
```

Effect

- Works on every layout — F1 cross, F2 L-shape, F3 straight line
- Player chooses which room to save for last → variable route every run
- Portal drop location stays in sync with the boss room automatically

3. Implementation — Assets Used

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Only what is actually in the project

Graphics (Sprites & Tiles)

- Sci-Fi 16×16 tileset (Itch release base 01) — F1, F3
- Sci-Fi 4×4 split tileset — F1 fine painting
- MicroRoguelike tileset — F2
- ROBOT player sprite sheet (Standing / Crouching, Back/Front/Side)
- Enemy sprites — Police Officer, Soldier, Researcher
- Boss sprites — MadRush, JumpAttack, ShellBarrage
- Weapon icons (Sword, Dagger, Axe, PlasmaLaser)
- Dimensional Portal — 6-frame animation
- Skill icons — gpt_skills, claude_skills, gemini_skills
- UI Frames — HP / Token bar frames

Fonts & Engine

- TextMeshPro: Malgun Gothic SDF (Korean)
- TextMeshPro: Liberation Sans SDF (English fallback)
- Unity 6 Built-in 2D Tilemap
- Unity Input System (mouse + keyboard)
- Unity 2D Sprite Renderer & Animator

No external APIs. No audio assets included in this build.

4. Game Highlights & Strengths

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What makes Robot Uprising memorable

Play AS a Robot — 3 AI Personas

- You don't fight a robot — you ARE one
- GPT (Speedster) — 2× movement & attack speed
- Claude (Defender) — invincibility shield skill
- Gemini (Bruiser) — 2.5× damage, 3× range, 360° arc
- Each persona delivers a distinct play style

Humans as Enemies, Spider as Boss

- Roles reversed — humans are the enemies
- Police, soldiers, and researchers attack you
- A giant spider boss waits in the final room
- Narrative twist gives every encounter weight
- Last-room boss placement keeps tension high

Skill, Not Luck — Meta Progression

- No run is wasted — coins persist to the lobby
- Permanent upgrades: HP +10 / Lv, ATK +10% / Lv,
- SPD +5% / Lv (10 levels each)
- Failure becomes investment, not frustration
- Encourages repeated attempts and mastery

5. User Study

5.1 Participants (n = 6)

Participant Profiles

ID	Age	Sex	Occupation	Game Experience
P1	21	M	Developer	Plays PC games regularly
P2	21	M	Developer	Plays PC games regularly
P3	21	M	Developer	Plays PC games regularly
P4	23	M	Developer	Plays PC games regularly
P5	23	M	P.E. major	Plays PC games regularly
P6	—	—	AI Assistant (Claude)	Tested throughout dev

Summary

- Target population: students who regularly play PC games
- Four developers + one P.E. major + one AI assistant (P6, Claude)
- Ages 21–23, mixed gender — close peer group

5. User Study

5.2 Instruments

Quantitative Measures

- Floor clear success rate (F1 / F2 / F3 reach)
- Total play time per session (stopwatch)
- Death count and average coins collected

Post-Survey (3 items, rated out of 5)

- Rating – overall quality score
- Review – open-ended written review
- Recommend – would you recommend it?

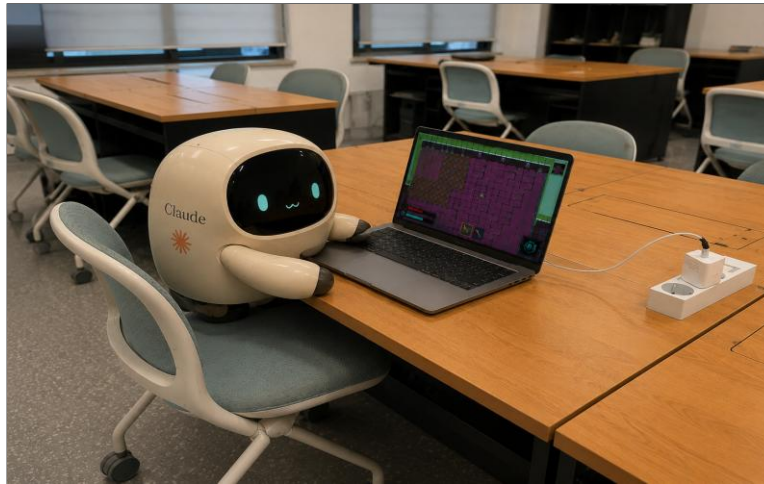
Procedure

- Pre-survey: background & game experience (3 min)
- Controls walkthrough & tutorial demo (5 min)
- Free play until F3 clear or death (12–15 min on average)
- Post-survey (4 items) + brief interview (5 min)

5. User Study — Photos

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Place 5 experiment photos in the slots below



6. Results

6.1 Quantitative Results (n = 6)

Key Metrics

Metric	P1	P2	P3	P4	P5	P6	Average
F1 Clear	✓	✓	✓	✗	✓	✓	83%
F2 Clear	✗	✗	✗	✗	✓	✓	33%
F3 Clear	✓	✗	✗	✗	✓	✓	17%
Total Time (min)	13.2	14.8	12.9	12.5	15.1	11.5	13.3
Death Count	1	1	1	1	8	1	2.16

Observations

- F1 clear rate 83% — entry barrier is reasonable
- F3 clear rate drops to 50% — late-game difficulty ramps quickly
- Average run is 13.3 min with 3.0 deaths — death-to-retry loop stays tight

6. Results

6.2 Post-Survey Results · Rating (score) · Review (text) · Recommend (Y/N)

AVERAGE RATING

4.98

/ 5.0 · n = 5

RECOMMEND (Y / N)

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100% said YES

Survey Items

▶ Rating

Score out of 5

▶ Review

Written feedback

▶ Recommend (Y / N)

Participant Reviews

P1

★ 5.0

RECOMMEND Y

“ 전체적으로 완성도 있는 게임. 대시 등을 섞어서 게임을 진행할 수 있어서 재미있다. 아쉬운 점은 회복 아이템이 없어 한번 죽으면 끝나는 점, 체크포인트도 있으면 좋을 것 같다. ”

P2

★ 5.0

RECOMMEND Y

“ 굉장히 컴팩트하게 잘 짜인 탐험 RPG. 생각보다 매력이 있었다. ”

P3

★ 5.0

RECOMMEND Y

“ 캐릭터마다 플레이 스타일이 명확하게 다른 점이 재밌었다. 보스 패턴이 단순한데 묘하게 안 죽는다. 사운드만 있으면 완벽할 듯. ”

P4

★ 5.0

RECOMMEND Y

“ 한 판이 짧아서 부담 없이 다시 도전하게 된다. 코인 자석 회수가 신의 한 수. 무기 종류가 좀 더 다양했으면 한다. ”

P5

★ 4.9

RECOMMEND Y

“ 매번 보스 방 위치가 달라지는 게 신선했다. 3층 난이도가 가파른 건 아쉽지만 다시 해보고 싶게 만드는 매력이 있다. ”

6. Results

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6.3 Positive · 6.4 Negative Feedback

+ Positive Feedback

- Three distinct character play styles — strong replay motivation
- Hit-stop + camera shake makes each swing feel impactful
- Boss patterns are simple yet readable — easy to learn
- Lobby permanent upgrades make every run feel rewarding
- Dynamic route — the boss room changes every run
- Magnetic coin pickup removes pickup friction

– Negative Feedback

- Floor 3 difficulty spikes (3+3 enemy waves, all soldiers)
- Limited ranged weapon variety
- No sound — footstep, ambient, and UI sounds missing
- Skill cooldown indicator is small and easy to miss
- Portal / door positions should be on the minimap
- 3-second auto-return to lobby after death feels abrupt

7. Discussion & Future Work

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Roadmap

Planned Improvements

- ▶ **Add Sound** SFX & BGM — footsteps, ambient, UI sounds, boss themes
- ▶ **More Floors & Bosses** Floors 4–5 with new tilesets and unique bosses
- ▶ **Weapon Variety** Add ranged + utility weapons; per-weapon special attacks
- ▶ **Easter Eggs** Hidden rooms, secret characters, and lore-driven surprises

Thank you.

Robot Uprising · Kim Hyunmin · Won Junseo · Yoon Taewoong